

The Brain.Space

Neuroscience-inspired cognitive system with 43 modules, 14 AutoGen agents, and 5 microservices.

Overview

The Brain.Space is Vibemind's knowledge discovery engine, built as a neuroscience-inspired system. It is NOT a traditional backend agent, but runs as a set of 5 standalone microservices with its own Docker stack.

Note: Brain does not have a backend agent in `python/swarm/backend_agents/__init__.py`. It is a standalone system in the Git submodule `python/spaces/brain/the_brain/` (github.com/Flissel/the_brain).

5 Microservices

Service	Port	Purpose
Brain Server	5000	Central unified brain + dashboards
Production API	5001	Legacy REST API
Swarm Server	5002	14 cognitive AutoGen agents
Memory API	8001	Supermemory integration
Dashboard	5000	Brain visualization

43 Neuroscience Modules

Inspired by real brain regions:

- **PFC** (Prefrontal Cortex) — Executive function
- **ACC** (Anterior Cingulate Cortex) — Conflict monitoring
- **Amygdala** — Emotional processing
- **VTA** (Ventral Tegmental Area) — Reward/motivation
- **Raphe** — Serotonin/mood regulation
- **Hippocampus** — Memory consolidation
- **Cerebellum** — Motor learning/prediction
- ... and 36 additional modules

9-Phase Cognitive Loop

Perceive → Appraise → Remember → Attend → Modulate → Reason → Reflect → Learn → Consolidate

14 Cognitive AutoGen Agents (Swarm Server)

The Swarm Server (port 5002) orchestrates 14 specialized agents via AutoGen for cognitive processing.

3D Visualization in the Multiverse

In `electron-app/renderer/multiverse.js`:

- Organic brain mesh (3D brain shape)
- 60 neuron particles with synaptic lines
- Position: (14, 0, -22), color: 0xff66aa (pink)
- Agent name: "Brain", role: "Knowledge Center"

Key Features

- **Evolutionary ML Pipeline**: Genetic algorithms for contextual transformation models (CTMs)
- **Semantic Clustering**: Dynamic knowledge clusters based on semantic noise
- **Continuous Learning**: Analysis of data streams for new connections
- **Pattern Discovery**: Automatic recognition of meaningful patterns
- **Intelligent Silence**: Learns when NOT to respond (anti-prompt injection)
- **Rowboat Integration**: Seeded integration with Rowboat data layer

Current Status

- 5 microservices architecturally defined and partially implemented
- 43 neuroscience modules present
- 9-phase cognitive loop implemented
- 14 cognitive AutoGen agents in the swarm server
- Evolutionary algorithm framework functional
- CTM training on semantic noise in research/validation phase
- Rowboat data integration started
- 3D visualization in multiverse active

Roadmap

- Complete CTM training methodology and parameter optimization (Q2-Q3 2026)
- Implement full continuous learning loop with Rowboat data
- Develop domain-specific clustering for common business scenarios
- Create visualization system for knowledge cluster exploration

- Add explainability layer showing why patterns were discovered
- Build prediction capability based on semantic cluster analysis
- Integrate with Ideas.Space for intelligent requirement mining
- Deploy production ML pipeline with multi-tenant isolation

Ecosystem Fit

The Brain.Space is the intelligence layer beneath Vibemind. Rowboat.Space feeds business data for analysis. Ideas.Space uses insights for requirements understanding. Desktop.Space learns user behavior patterns. Coding.Space accesses recognized patterns. The entire platform becomes progressively more intelligent through the connections that The Brain discovers.